

What is Software Quality Assurance?

Let's break it down into three parts:

1. Software

You already know this - it's the programs and applications we use on computers, phones, etc.

2. Quality

Think of quality as "how good something is." For example:

- When you buy a phone, you check if it works well, has good battery life, and doesn't crash - that's checking its quality
- If you're comparing two restaurants, you look at the food taste, service, and cleanliness - you're comparing their quality

Important point: You compare similar things! You wouldn't compare a car with a bicycle to determine quality, right? You compare a car with another car.

3. Assurance

This simply means a "promise" or "guarantee." Like when a company says "We assure you this product will work for 2 years" - they're making a promise.

So, Software Quality Assurance =

A promise that the software is good and will work as expected!

The Big Picture: SQA vs Testing

Here's where students often get confused:

Testing is like checking your exam paper before submitting it - you're looking for mistakes.

SQA is much bigger! It's like:

- Planning how to study (requirements)
- Organizing your notes properly (design)
- Actually studying (development)
- Checking your understanding (code reviews)
- Taking practice tests (testing)
- Reviewing what went wrong (maintenance)

In other words: Testing is just ONE part of SQA. SQA covers the ENTIRE process of making software, from start to finish!
